

ANIRUDH P

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI Engineer-focused Computer Science (AI & ML) student with hands-on experience building end-to-end ML systems — from computer vision pipelines (YOLOv8, EasyOCR) to Retrieval-Augmented Generation (RAG) applications using LangChain, vector databases, and open-source LLMs. Experienced in deploying models as containerized, production-style APIs (Fast API, Docker) and integrating LLMs with real-world structured data. Strong foundation in IoT and hardware - software integration.

TECHNICAL SKILLS

- **AI & ML:** Computer Vision (YOLOv8, EasyOCR), Retrieval-Augmented Generation, LangChain, ChromaDB, Sentence-Transformer Embeddings, Deep Learning
- **LLM & APIs:** Groq (LLaMA 3), OpenAI/Anthropic APIs, Prompt Engineering, REST API Design (FastAPI, Flask)
- **Languages:** Python, JavaScript (ES6+), SQL, C++, Java, Kotlin
- **Deployment & Tools:** Docker, Git/GitHub, Linux, VS Code, Uvicorn
- **Web & Mobile:** HTML5, CSS3, React, Node.js, Kotlin & SQLite
- **IoT & Electronics:** ESP32, Arduino Uno, Sensor Integration, Circuit Design, Firmware Development

KEY PROJECTS

Traffic Violation RAG Assistant — LangChain, ChromaDB, Groq LLaMA 3, FastAPI, Docker

- Built a Retrieval-Augmented Generation system enabling natural-language queries over a 250-record traffic violation database, using sentence-transformer embeddings (all-MiniLM-L6-v2), a ChromaDB vector store, and Groq-hosted LLaMA 3 for grounded, context-aware answer generation.
- Achieved over 90% top - 4 retrieval relevance on natural-language test queries, with end-to-end query response time under 2 seconds using Groq's low-latency inference.
- Designed and deployed a FastAPI service with /query and /ingest endpoints, fully containerized with Docker for reproducible one-command deployment, extending an earlier YOLOv8 detection pipeline into a conversational data interface.

Traffic Violation Detection System — YOLOv8, EasyOCR, Flask

- Engineered an automated pipeline using YOLOv8 and EasyOCR to detect vehicles and recognize license plates in real-time from camera feeds.
- Developed a centralized Flask dashboard to visualize live detection data and manage historical violation logs.

Smart Soil Health & Compost Monitor — ESP32, IoT

- Architected an IoT system using ESP32 to monitor soil pH, moisture, and temperature via integrated hardware sensors, with cloud-based alert logic to optimize composting efficiency through real-time remote tracking.

Expense Tracker Mobile Application — Kotlin, SQLite

- Developed a native Android application using Kotlin for daily financial tracking and budget management, with SQLite for secure local persistence of transaction records.

CERTIFICATIONS

- Ethical Hacking — Udemy (March 2026) • Deep Learning — Simplilearn (May 2025)
- Mastering Retrieval-Augmented Generation (RAG) — Udemy (April 2025) • Learn Python Masterclass — Udemy (March 2025)

EDUCATION

BCA (AI & ML) — Presidency University, Bangalore Jun 2023 – May 2026 (Exp.)

PCME — St. Joseph's PU College, Bangalore Aug 2021 – Mar 2023